Y. Harsha Vardhan Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.85
Sri Chaitanya Junior College	Class XII (Telangana State Board)	2024	97.80%
Sri Chaitanya Techno School	Class X (Telangana State Board)	2022	10.00%

SCHOLASTIC ACHIEVEMENTS

- Received A+ Grade in 3 courses (Awarded to top 2% of the class)

PROJECTS

Dual-Core SIMT GPU | GPU CUDA Architecture | GitHub

- Designed a **Dual-Core** SIMT GPU with a **Dispatcher** and **Queue-Based Memory Scheduler** for handling Memory-requests.
- Built a per-core Fair **Round-Robin** Warp Scheduler with zero-cycle latency switching across 4 Warp Slots, 16 Threads each.
- Designed a **Custom ISA** and allowed each warp to execute Independent Instruction Streams from separate regions of I-Mem.
- Handled **Branch-Divergence** using a **Mask-Based** Execution Model with an 8-Entry **Divergence Stack** .
- Validated the design with **MatrixAddn/Multiply Kernels**, deployed on PYNQ-Z2 at **25MHz** using **28.4k LUTs** with **interrupt**-based done signaling.

FPGA Based Real-Time Video Processing Pipeline | Peripheral Interfacing | GitHub

- Built an **OV7670 camera capture pipeline** (640*480) with a custom **SCCB Driver(I2C Like)** for camera initialization.
- Generated **VGA display timing** at (800*600) at 60Hz with at a clock speed of 40MHz and used RGB-to-DVI IP core.
- Handled **CDC** between 24MHz and 40MHz Clock domains using **Dual-Port BRAM** with **Blanking-Interval Technique**.
- Implemented a **Streaming Video Pipeline** using **Line Buffers** to exploit Temporal and Spatial **Locality** across frames.
- Architected Pipelined **2D Convolution** Kernels for filters like Sobel Edge Detection, with an AXI-Stream FIFO interface for standard-compliant **Back-Pressure** handling.
- Deployed and tested on PYNQ-Z2 FPGA via **PMOD interfaces**, with HDMI Monitor at (800*600) for Output validation.

CNN Accelerator Based MNIST Classifier in Verilog | Hardware-Accelerator, Parallel-Processing | GitHub

- Architected a RTL Design for **High-Throughput** Convolutional Neural Network (CNN) accelerator from scratch
- Implemented the design in Verilog to continuously s**tream, process,** and **classify** handwritten digits from the MNIST dataset.
- Designed a circular line-buffer to reconstruct **3×3 pixel windows** from live serial pixel stream, **avoiding full-frame buffering**.
- Fed the reconstructed windows into **8 parallel 7-stage** pipelined convolution cores.
- Each core used **optimized adder trees** for convolution and **2×2 max-pooling** modules for downsampling.
- Designed FC digit neurons in a Q(I,F) **Fixed-Point** format, executing 1,352 streaming **MAC Operations** per classification.

5-Stage Pipelined Parameterizable RISC-V Processor | Verilog, Vivado, AXI, PYNQ-Z2 FPGA | GitHub

- Designed a 5-stage pipelined RISC-V core with configurable 32/64-bit datapath width and modular RTL architecture.
- Implemented **Hazard Detection** and **Data Forwarding** to manage **RAW Hazards** across the pipeline.
- Built a **2-Bit Saturating Branch Predictor** with an **8-Entry Branch-History Table (BHT)** for branch prediction.
- Designed an **FSM-based BRAM interface** to stall the pipeline during memory access latency.
- Deployed on **PYNQ-Z2** using **AXI** to load programs and read back results via the PS, an **interrupt-driven** done

signal.
- Verified functionality through simulation and on-board testing in Xilinx Vivado.

SKILLS

- **Hardware Design**: RTL Design | FPGA Design | Video-Pipeline Interfaces | Peripheral Interface Design
- **Computer Architecture** : CUDA GPU Architecture | RISC-V Architecture | CNN Accelerator Architecture
- **Digital Design Tools** : Xilinx Vivado | Intel Quartus Prime | Icarus Verilog | GTKWave | PYNQ-Z2 (Zynq-7020) | cocotb
- **Analog Design Tools** : KiCad | Altium Designer | LTspice | NGSpice
- **Programming Languages**: Verilog | C | Python | MATLAB | RISC-V Assembly | LaTeX

RELEVANT COURSES

- **Digital Design & Labs**: Computer Architecture | Digital Systems Lab | Circuits Lab | Electronic Devices and Circuits Lab | DSP-Lab (Implemented Filters in Verilog) | PCB Design and Layout for Electronic Systems
- **Electronics**: Circuits and Network Analysis | Digital Systems | Analog Electronics | Engineering Electromagnetics
- **Mathematics & Programming**: Probability | Machine Learning | Intro to Programming | DSA

POSITIONS OF RESPONSIBILITY AND EXPERIENCE

Overall Head | Elektronika - Electronics and Signal Processing Club, IIT Hyderabad

- Led club operations and technical direction, managing hardware projects and organizing technical community-sessions.
- Driven end-to-end execution of hardware projects, handling design workflows, resolving technical bottlenecks with teams.

Teaching Assistant | Circuits and Network Analysis (EE1101)

- Conducted doubt-solving sessions to guide students for improving their clarity and understanding of the coursework.
- Led sessions on **NGSpice** for circuit simulations, designing practical assignments and evaluating student performance.

EXTRACURRICULAR

- Hobbies**:** Drawing, watching cricket, playing cricket, playing badminton and an NSS Volunteer.